How Variations of the Sierra Nevada Snowpack Size Affects Lake Tahoe Water Quality

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Introduction

Lake Tahoe is the largest alpine lake in North America, on the California-Nevada state border. Its remarkably clear waters support a multi-billion dollar recreation economy, critical fisheries, and serve as a crucial drinking-water reservoir for cities in Nevada (Toy, 2018) Sitting above 6,000 feet, a significant water source of Lake Tahoe is runoff from the melted snowpack of the Sierra Nevadas. Every spring, this runoff supplies new water to the lake through rivers like the Upper Truckee. The magnitude and timing of the snowmelt of this event varies dramatically from year to year, which has relatively well-documented consequences for downstream water management. Less known, however, is what such inter-annual variability means for the lake's water quality (Wang et. al, 2020). Snowmelt simultaneously dilutes existing nutrients and sediments, while bringing in new loads of matter, meaning the *net* effect on water quality is not straightforward. As California droughts become increasingly unpredictable, understanding this relationship will become critical for ecological management and public health decisions (Swain et. al, 2018). We are interested in determining: ***How do inter-annual variations in the magnitude and timing of the Sierra Nevada snowpack melt affect the turbidity and chlorophyll content of Lake Tahoe from 2017 - 2025?***

Data

Sentinel 2: Harmonized Sentinel 2 MSI Level-2A Surface Reflectance

Lake Tahoe Shapefile: A GEOJSON file that outlines the entirety of Lake Tahoe

USGS Hydrographs: Graphs of the amount of water flow going into Lake Tahoe from each year

SNOTEL: Snowpack telemetry that measures snowpack in Sierra Nevada range. Snowpack measured in snow water equivalent

Methods

Figures & Visualization

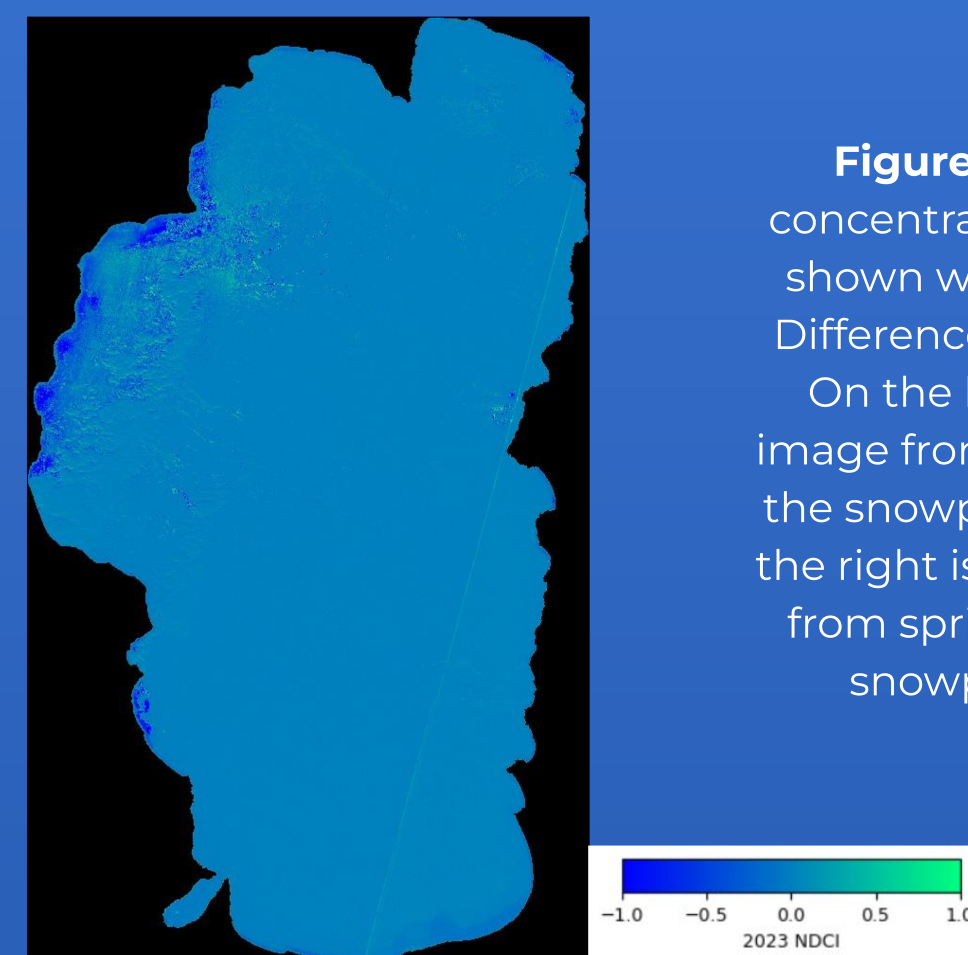
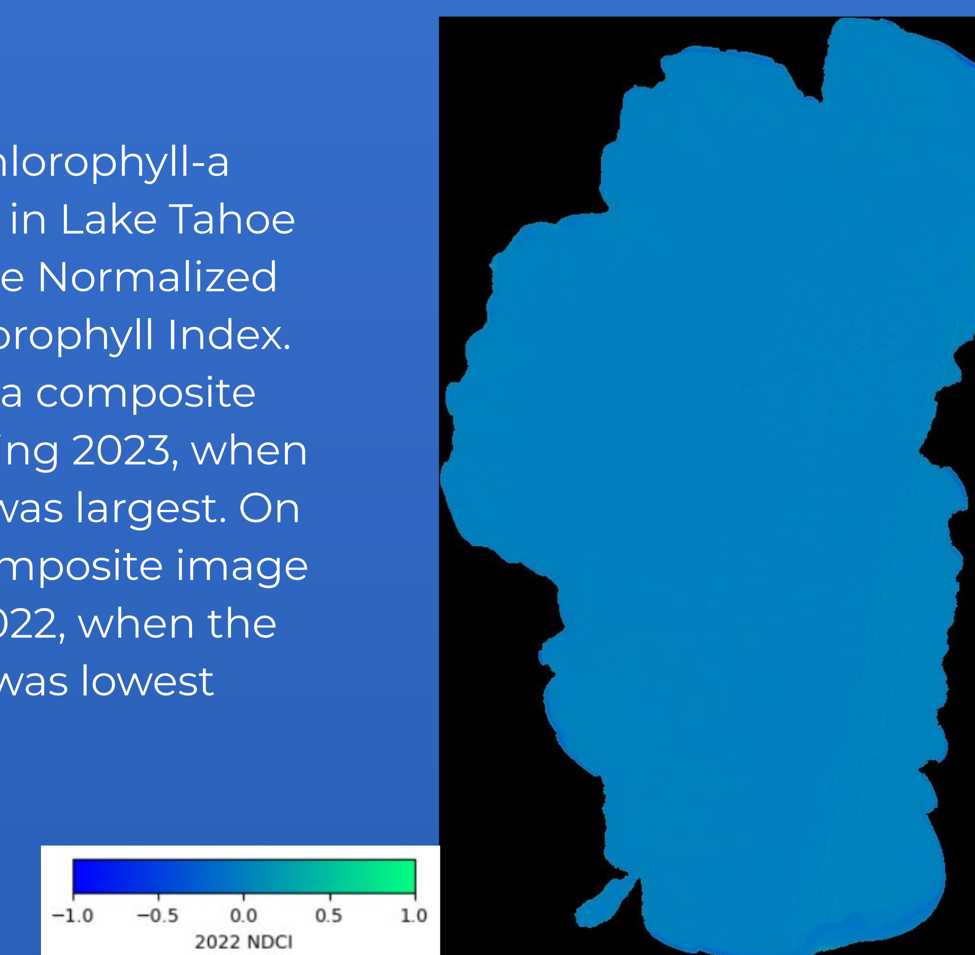


Figure 1: Chlorophyll-a concentrations in Lake Tahoe shown with the Normalized Difference Chlorophyll Index. On the left is a composite image from spring 2023, when the snowpack was largest. On the right is a composite image from spring 2022, when the snowpack was lowest



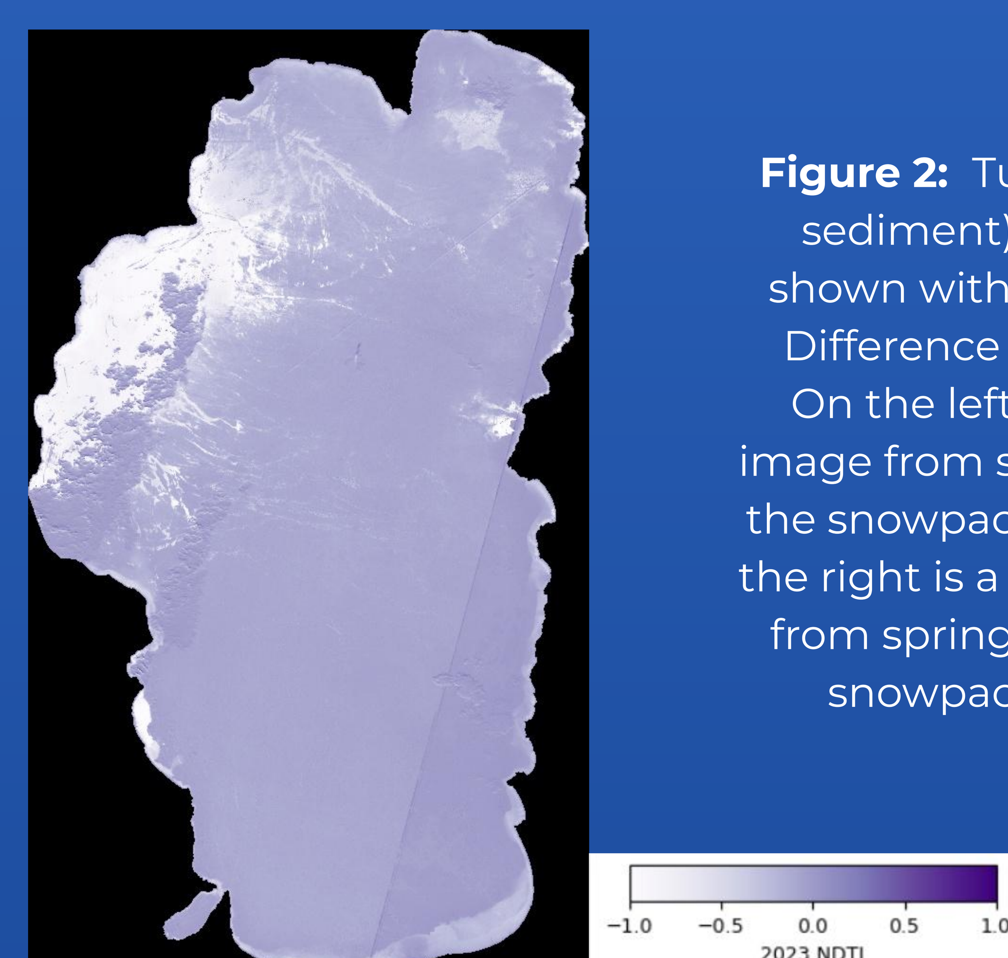
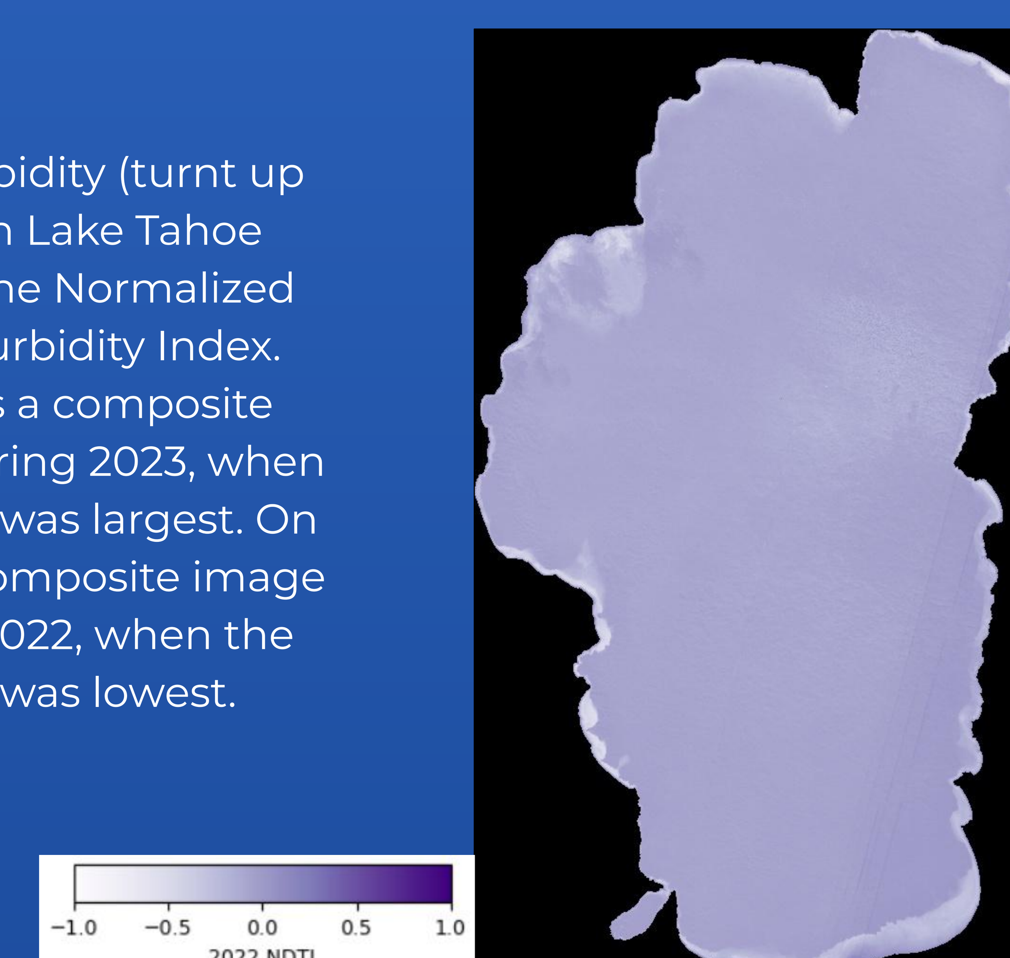
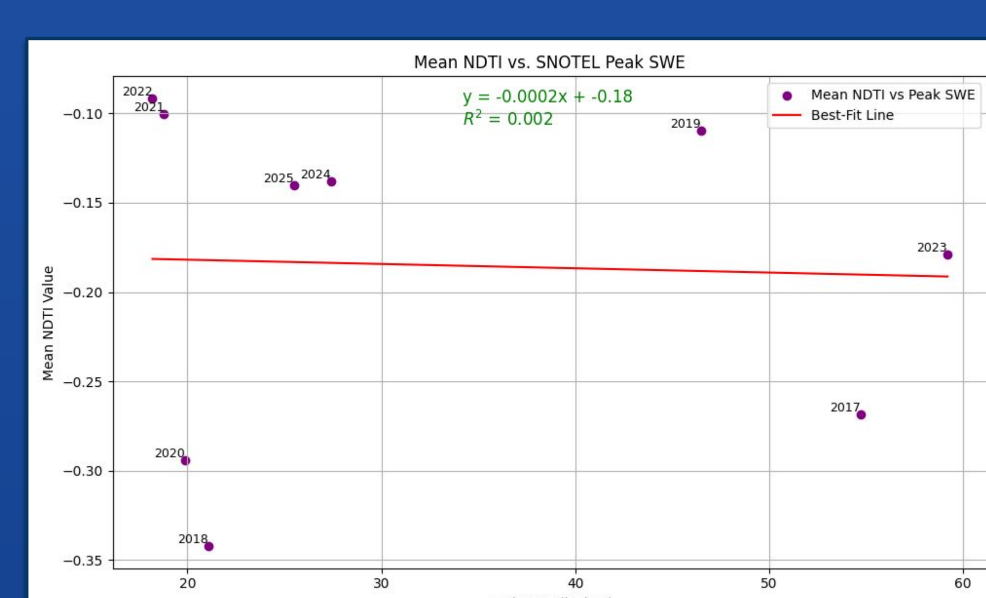


Figure 2: Turbidity (turnt up sediment) in Lake Tahoe shown with the Normalized Difference Turbidity Index. On the left is a composite image from spring 2023, when the snowpack was largest. On the right is a composite image from spring 2022, when the snowpack was lowest.





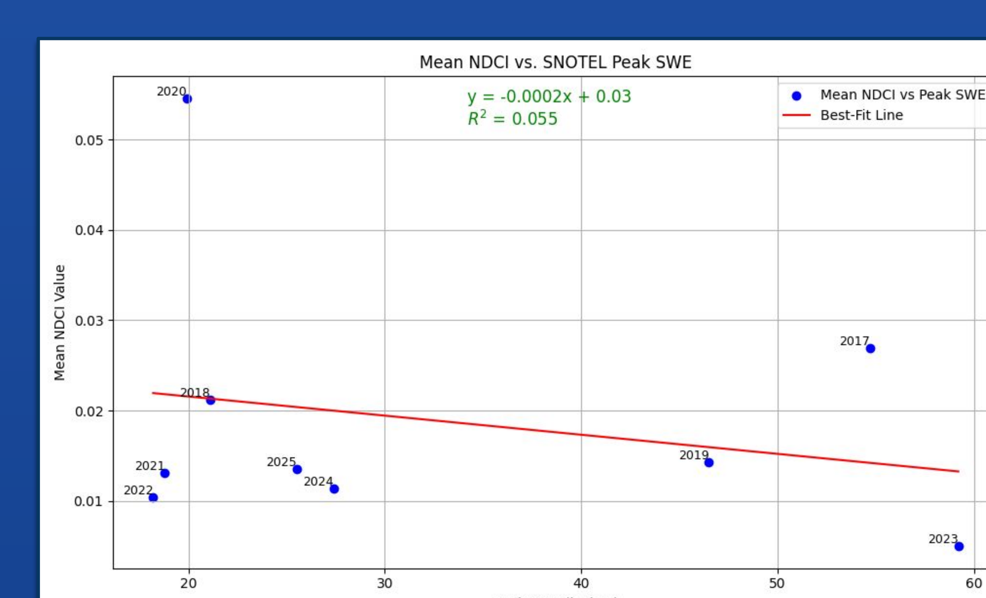


Figure 3:
Left: linear regression of the relationship between turbidity (y-axis) and peak snowpack levels (x-axis).
Right: linear regression of the relationship between chlorophyll-a (y-axis) and peak snowpack levels (x-axis)

Results

The R^2 value for average NDCI and snowpack size was 0.002 between 2017 and 2025. The R^2 for average NDTI and snowpack size for the same timeframe was 0.055. Therefore there is no relationship between snowpack size and both chlorophyll concentration and turbidity in Lake Tahoe. Additionally, we found that because both overall NDCI and NDTI levels are low (ranging from -0.35 to 0.06), our analyses finds that Lake Tahoe has very clean water.

Discussion

There is no strong relationship between snowpack size and chlorophyll or turbidity in Lake Tahoe. This does not support our hypothesis that more snowmelt improves water quality due to dilution from the pure water influx. It points to the idea that the quantity of snowmelt entering Lake Tahoe is not a factor for water quality, or at least chlorophyll-a concentrations and turbidity.

Limitations

Although the water on the western side of the lake appears very clear according to the NDTI, this is because of photosynthetic growth reflecting more green light than sediment is reflecting red light, making the NDTI (Red-Green / Red+Green) negative. So the water likely has low sediment concentrations, but is not pure because there is potential algal growth.

Future Work

Future extensions would incorporate in-situ sediment and turbidity data to validate our NDCI and NDTI, expand independent variables to include thermal imagery, and apply a multivariate regression to better isolate the drivers of water quality variations. Later, as Sentinel-2 imagery accumulates, we can expand the time series to potentially enhance inter-annual patterns. Ultimately, these steps may build toward an AI-powered "digital twin" of Lake Tahoe's water system; a fusion of satellite imagery, continuous in-situ sensing, and physics-informed hydrology models into a dynamic, virtual replica of the lake. This would enable real-time water quality monitoring, improved algal bloom prediction, and proactive management decisions amongst intensifying drought variability.

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